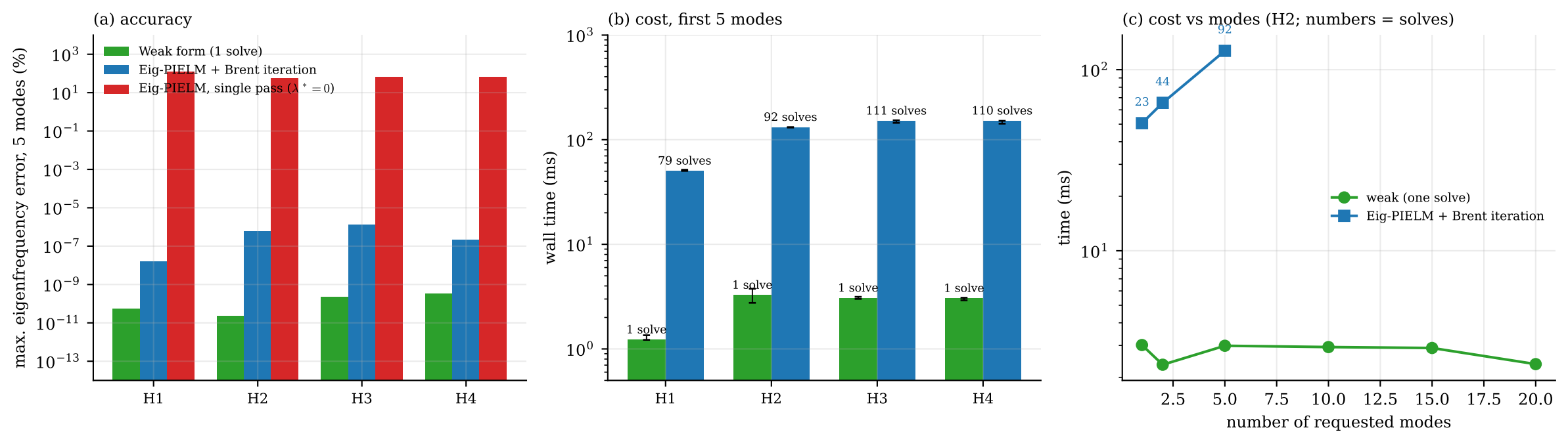
(a) accuracy
Weak form (1 solve)
Eig-PIELM + Brent iteration
Eig-PIELM, single pass ($\lambda^* = 0$)
max. eigenfrequency error, 5 modes (%)
10^{3}
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
H1
H2
H3
H4
(b) cost, first 5 modes
wall time (ms)
10^{3}
10^{2}
10^{1}
10^{0}
79 solves
92 solves
111 solves
110 solves
1 solve
1 solve
1 solve
1 solve
H1
H2
H3
H4
(c) cost vs modes (H2; numbers = solves)
time (ms)
10^{2}
10^{1}
92
44
23
weak (one solve)
Eig-PIELM + Brent iteration
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
number of requested modes